



Hacktux User Manual

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1. Introduction

1.1. Purpose of the Manual

This manual is designed to provide comprehensive instructions on how to effectively use Hacktux application on your computer. It aims to assist users in understanding the functionalities and features of the software, ensuring a seamless user experience.

1.2 Scope

The manual covers the installation, setup, and usage of Hacktux application. To carry out its tasks, Hacktux needs to interact with another piece of software called VMware Workstation Pro. This manual will also briefly show you how to install this software and interact with it in a basic way. If the user needs or wishes to acquire more specific knowledge about VMware Workstation Pro, the VMware Workstation Pro user manual for the recommended version is available together with this manual.

1.3 Target Audience

This manual is intended for new users and existing users who seek guidance Hacktux functionalities. It assumes a basic understanding of computers, software usage, and knowledge of virtualization technologies and functionalities.

1.4 About Hacktux

Hacktux is a computer program that provides its users with Cyber Ranges related to ICS/OT area. To do this, it uses VMware Workstation Pro as external software, which allows it to virtualize virtual machines that will be used to create the virtual environment where the user can carry out attacks to increase their knowledge of cybersecurity in the ICS/OT area.

Hacktux is free and open source, so users can contribute to the evolution of the application if they wish. VMware Workstation Pro can also be used for free, for this you need to use version 17.5.2 or higher, this version has been updated to make VMware Workstation Pro available for free if it is used for personal use, it is important to note that versions prior to this will require licenses which will force the user to purchase one, in addition Hacktux has been tested and designed to work with version 17.5.2 for this reason earlier versions may not work correctly with Hacktux.

2. Getting Started

2.1. System Requirements

Before proceeding with the installation, ensure that your system meets the following requirements:

- Operating System: Windows 10, Windows 11
- Processor: An Intel CPU compatible with VT-x support, An AMD CPU compatible with AMD-V support
- RAM: 8 GB or higher
- Disk Space: 70 GB or higher

2.2. Installation Instructions

Before installing Hacktux you first need to install VMware Workstation Pro, version 17.5.2, follow these steps to install VMware Workstation Pro.

2.2.1 Install VMware Workstation Pro, version 17.5.2

WARNING: The following guide to installing VMware Workstation Pro, version 17.5.2, was tested on 06/09/2024, if any step does not work correctly, or if any step can no longer be performed for some reason, we advise you to investigate on the internet what is the most recent way to install VMware Workstation Pro, version 17.5.2.

1 - Search on the internet for "download vmware workstation pro" or click on this link <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

2 - After entering the website press the button that says "Download Fusion or Workstation"

Desktop Hypervisor

Run Windows, Linux and other virtual machines with VMware Workstation Pro for Windows and Linux or VMware Fusion for Mac, the industry standard desktop hypervisors.

DOWNLOAD FUSION OR WORKSTATION

SUBSCRIBE ONLINE

Figure 1 - Button to download VMware Workstation Pro

3 - After pressing the button, you will be redirected to a "Broadcom" webpage which will ask you to log in. If you already have an account, enter the credentials and go to step 10, if not continue to step 4.

Having trouble logging in? [Click here](#) to use our Chat Bot for assistance.

Connecting to Support Portal ECX

 **BROADCOM®**

Broadcom Inc. Customer sign-in

Username

Enter your username

☐ Remember me

Next

Figure 2 - Login Page for Broadcom Portal

4 – If you don't have an account on Broadcom, go to the top right and press on login, after that press on "Register"

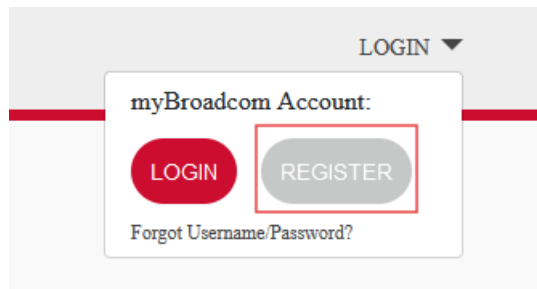


Figure 3 - Place where you need to press to create an account

5 – After entering on the registration page put your email, do the challenge and press on the button “Next”

 A screenshot of the Broadcom Support Portal User Registration page. The page has a grey background with a geometric pattern. On the left, the text 'Broadcom Support Portal' is displayed in blue, followed by the message 'Enhance your skills through multiple self-service avenues by creating your Broadcom Account'. On the right, there is a white box titled 'User Registration' with the subtitle 'Create your Account'. Inside this box, there is a form with the following elements:

- 'Email Address' label followed by a text input field containing 'name@company.com'.
- A red error message below the input field: 'Email is required'.
- A CAPTCHA challenge showing a distorted image with the letters 'RJCB' overlaid. Below the image is a text input field labeled 'Enter text from image'.
- A blue 'Next' button at the bottom of the form.
- A warning message below the button: '⚠ Use of a shared email account or a distribution list (PDL) is a security risk and a violation of the Terms of Use for this Site.'
- A link at the bottom: 'Having trouble? Ask our chatbot for assistance.'

Figure 4 - Place to enter email

6 – After pressing on the “Next” button you will need to go to your email to put the code that you received, then press the button to “Verify & Continue”

The screenshot shows the 'Broadcom Support Portal' header on the left with the text 'Enhance your skills through multiple self-service avenues by creating your Broadcom Account'. On the right, a white modal box titled 'Verify your Email Address' contains a back arrow, a title, a label 'Enter the verification code sent to', a single text input field, and a six-digit numeric keypad. Below the keypad are links for 'Resend' and 'Click here for help', and a 'Verify & Continue' button at the bottom.

Figure 5 - Place to enter the code you received in email

7 – After pressing on the “Verify & Continue” button you will need to fill the form with your data so that the account can be created, after fill all form with your data press “Create Account”

The screenshot shows the 'Broadcom Support Portal' header on the left. On the right, a white modal box titled 'Complete your Registration' contains a back arrow, a title, and several form fields: 'First Name' and 'Last Name' (text inputs), 'Country' and 'Job Title' (dropdown menus), 'Password' and 'Confirm Password' (text inputs). At the bottom, there is a checkbox for 'I accept the Terms of Use' with links to 'Terms of Use' and 'Privacy Policy', and a 'Create Account' button.

Figure 6 - Form that you need to fill with your data, to create an account

8 – After pressing on the “Create Account” button you will be redirected to a page that asks you if you want to build your profile, that’s not necessary for installing VMware so because of that, press “I’ll do it later”

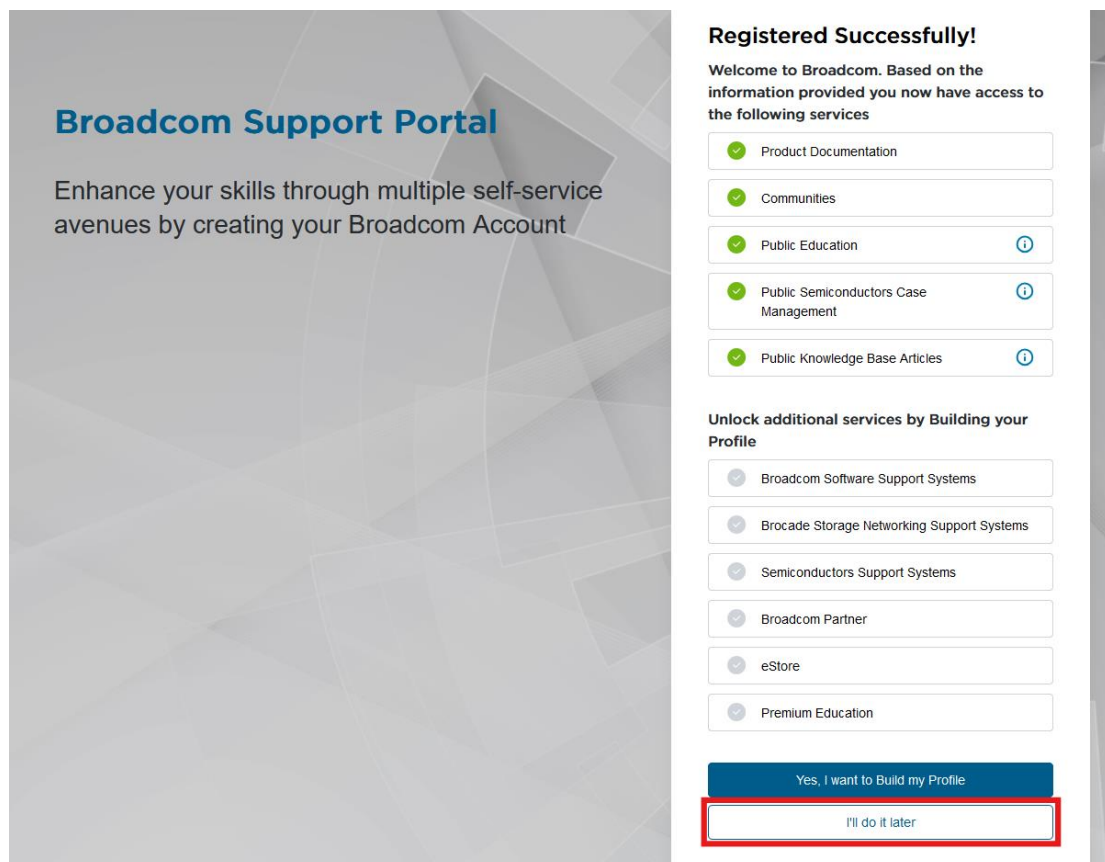
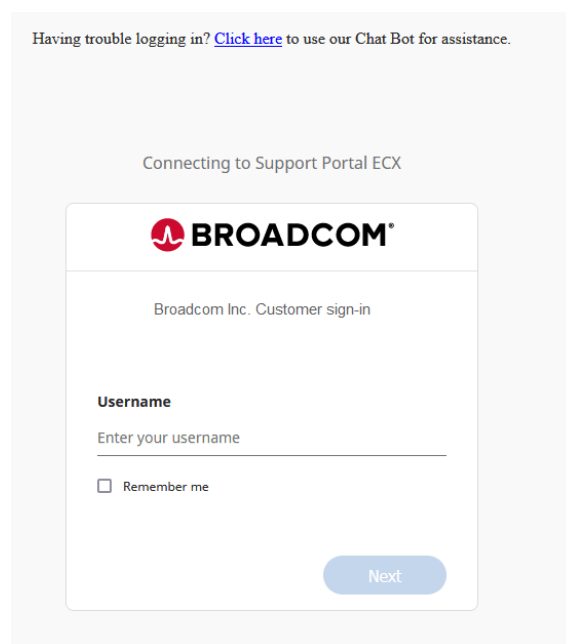


Figure 7 - Place where you can build your profile or do it later

9 – Once the account has been successfully registered, you should return to the login page obtained after step 2, here is the link <https://support.broadcom.com/group/ecx/downloads>



Having trouble logging in? [Click here](#) to use our Chat Bot for assistance.

Connecting to Support Portal ECX

BROADCOM®

Broadcom Inc. Customer sign-in

Username

Enter your username

☐ Remember me

Next

Figure 8 - Login page to Broadcom Portal

10 - After logging in, the user will be redirected to the Broadcom portal

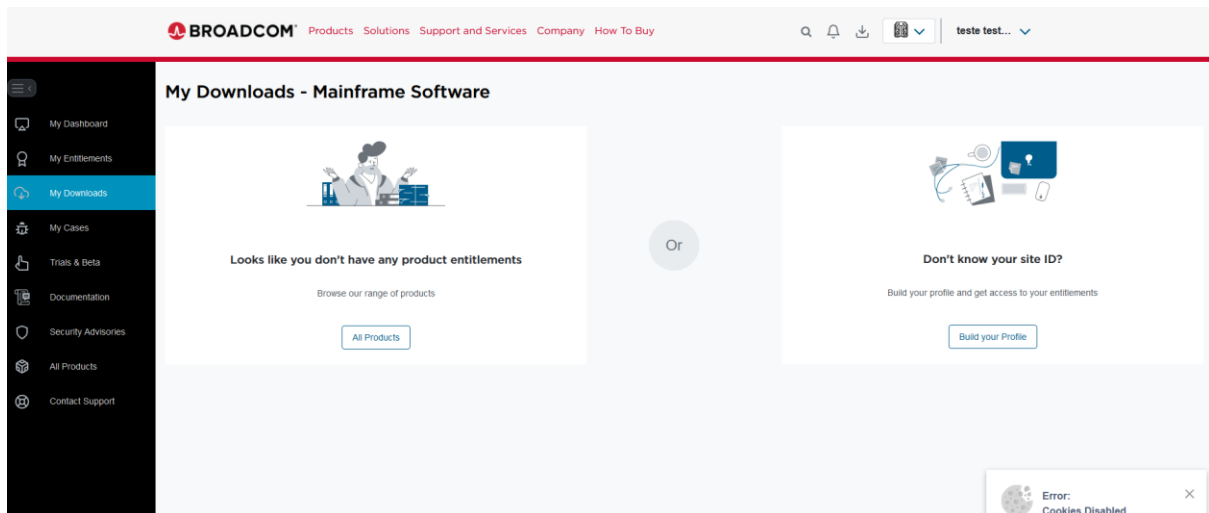


Figure 9 - Portal of Broadcom

11 - After entering the portal, you must first go to the checkbox next to your username on the right-hand side and select the "VMware Cloud Foundation" option.

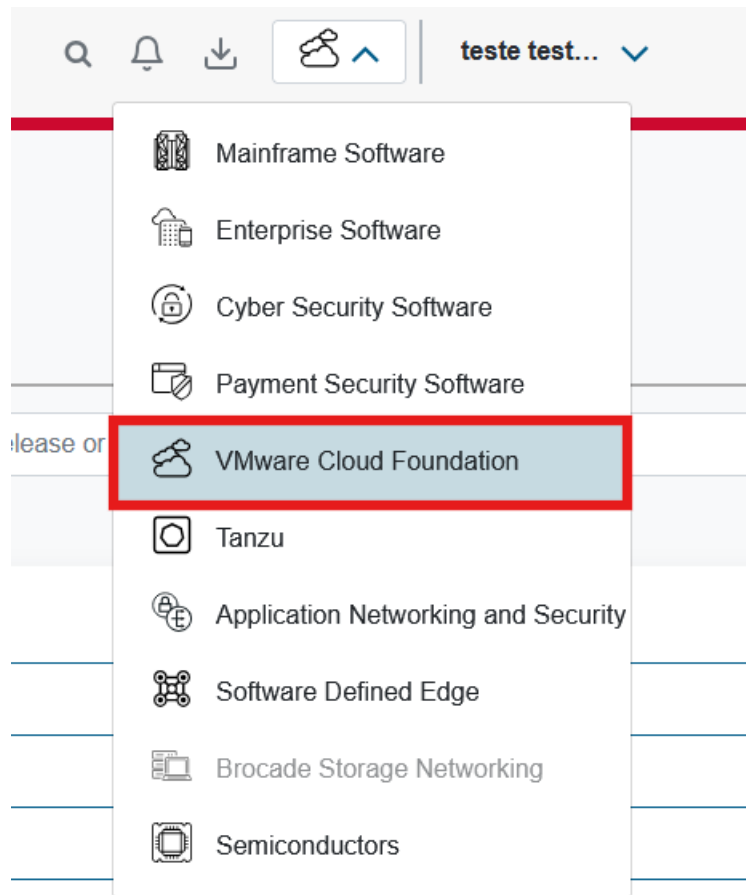


Figure 10 - Option that you need to choose on the checkbox

12 - After that, the user should go to the "My Downloads" section, which will be updated to show VMware software, and so there you just must look for the button that says, "VMware Workstation Pro".

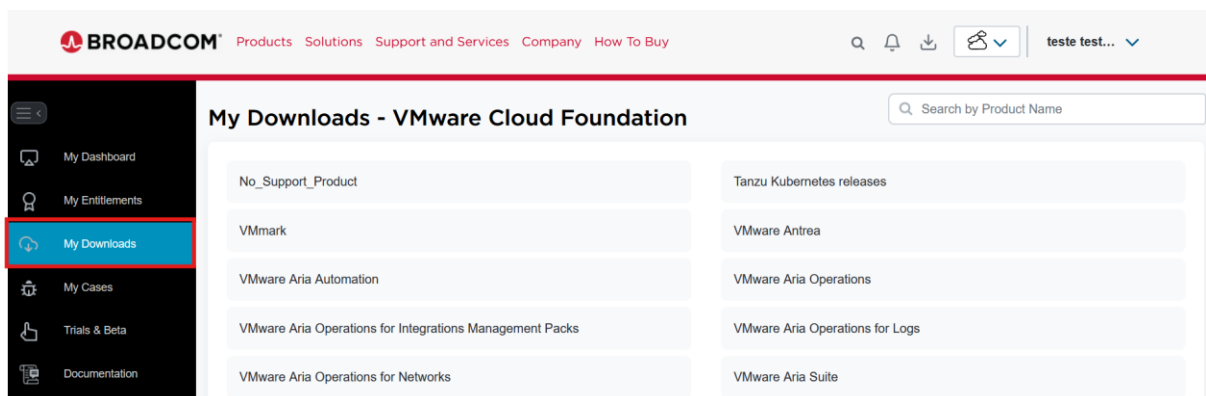


Figure 11 - My Downloads section fill with VMware products

VMware Live Recovery	VMware NSX
VMware NSX Data Center for vSphere	VMware NSX Intelligence
VMware NSX-T Data Center	VMware Photon Platform
VMware Site Recovery Manager	VMware Skyline Collector
VMware Tools	VMware Validated Design for Software-Defined Data Center
VMware Workstation Player	VMware Workstation Pro

Figure 12 - List of VMware products and the VMware Workstation Pro software

13 - After entering the VMware Workstation Pro section, simply search for version 17.0 for personal use, version Windows, remember that you should choose version 17.5.2 because to date this is the only version that supports using VMware for personal use.

The screenshot shows the Broadcom VMware Workstation Pro product page. The left sidebar contains navigation links: My Dashboard, My Entitlements, My Downloads, My Cases, Trials & Beta, Documentation, Security Advisories, All Products, and Contact Support. The main content area is titled 'VMware Workstation Pro' and includes a search bar. Below the search bar, a list of products is displayed. The product 'VMware Workstation Pro 17.0 for Personal Use (Windows)' is highlighted with a red box.

Figure 13 - List of products, VMware Workstation Pro

The screenshot shows the Broadcom VMware Workstation Pro product page, specifically the 'Products' section. The product 'VMware Workstation Pro 17.0 for Personal Use (Windows)' is selected, and a table of versions is displayed. The version '17.5.2' is highlighted with a red box.

Release	Release Level Info	Language
17.5.2	520448	English
17.5.1	520407	English
17.5	520406	English
17.0.2	520405	English
17.0.1	520404	English
17.0.0	520403	English

Figura 14 - List of versions, VMware Workstation Pro for Windows for personal use

14 - When you enter the VMware Workstation Pro (for windows) version 17.5.2 page, you need to accept the terms and press the executable download icon.

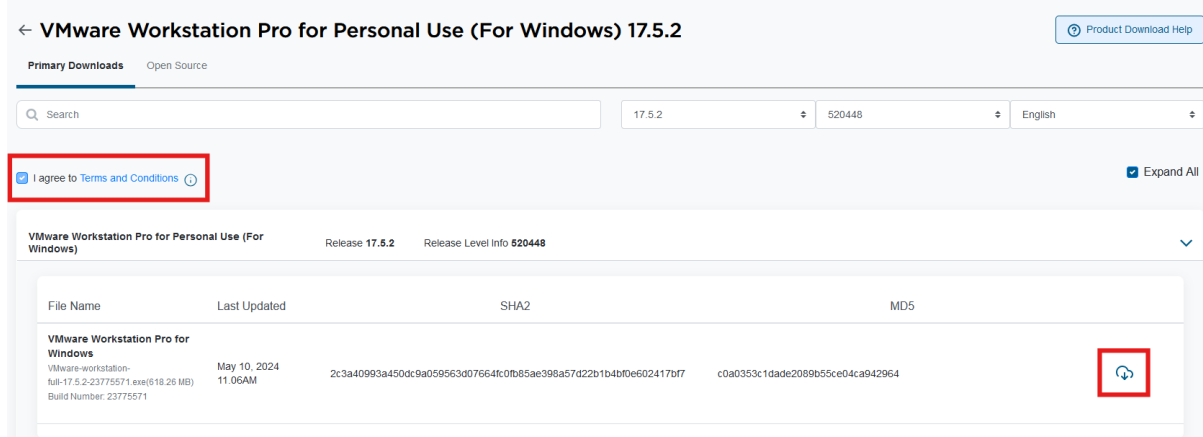


Figure 15 - Place where you can download VMware Workstation Pro for windows version 17.5.2

15 - After pressing the download icon, a message will appear indicating that further checks need to be made, click on the "yes" button.

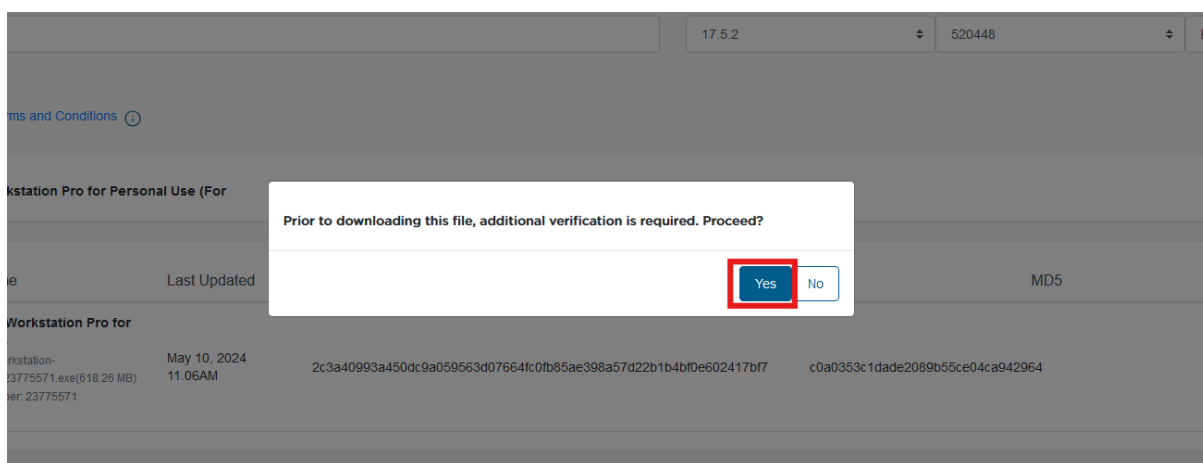


Figure 16 - Additional verification box

16 - The extra check you need to make is to enter the user's address details, after all you must click on the "Submit" button

Trade Compliance Verification

First Name Last Name Email Company

* Address1 Address2 * City * State/Province

* Country * Zip/Postal Code

☒ I Agree ☐ I Do Not Agree

I understand that the export laws and regulations of the United States and other countries apply to the use, export, and re-export of this software (including associated documentation, technical information, and support, "Software"). I will not use, export, or re-export the Software, directly or indirectly, except as permitted by United States and other applicable law. In particular, but without limitation, I will not export or re-export the Software, directly or indirectly, (i) into (or to a national or resident of) any U.S. embargoed country (ii) to anyone on the U.S. Department of Commerce Table of Denial Orders or the U.S. Treasury Department list of Specially Designated Nationals, or Specially Designated Terrorists, or Specially Designated Narcotic Traffickers; or (iii) for any end use that is prohibited by United States or other applicable law. I represent that I am not located in, under control of, or a national or resident of any such country or on any such list and that the U.S. government has not suspended, revoked, or denied my export privileges. I affirm that I will not use the Software for nuclear, missile, chemical biological weapons, or other weapons of mass destruction. (For more information on the U.S. export laws, see the Bureau of Industry and Security's home page.)

Figure 17 - Address information space

17 - After filling in the information related to the user's address, click on the download icon of VMware Workstation Pro page to download the executable.

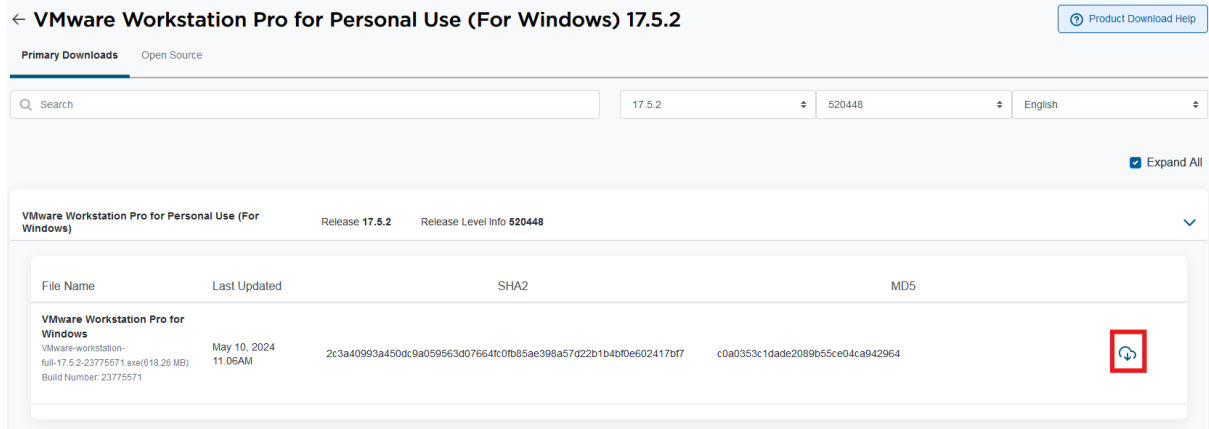


Figure 18 - Place to download VMware Workstation Pro

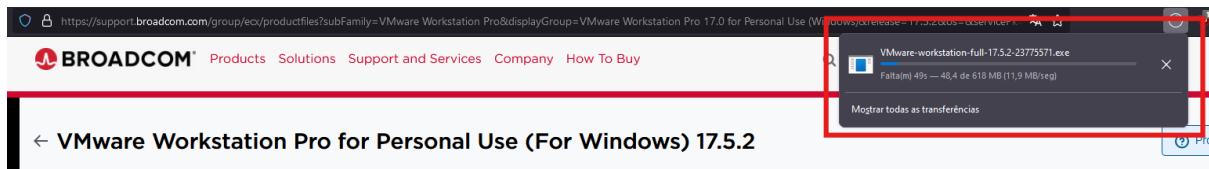
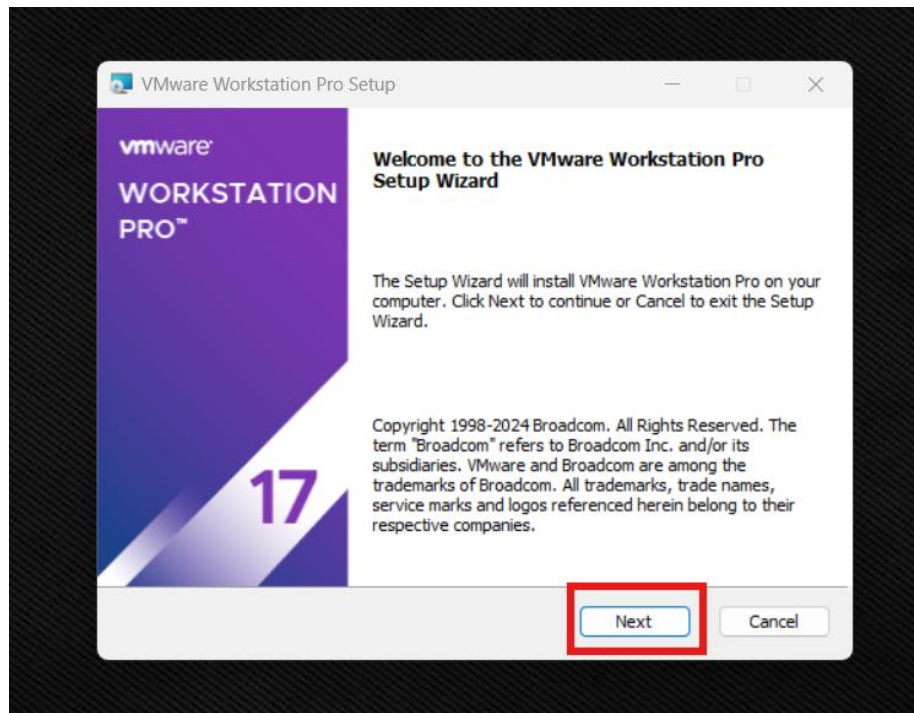
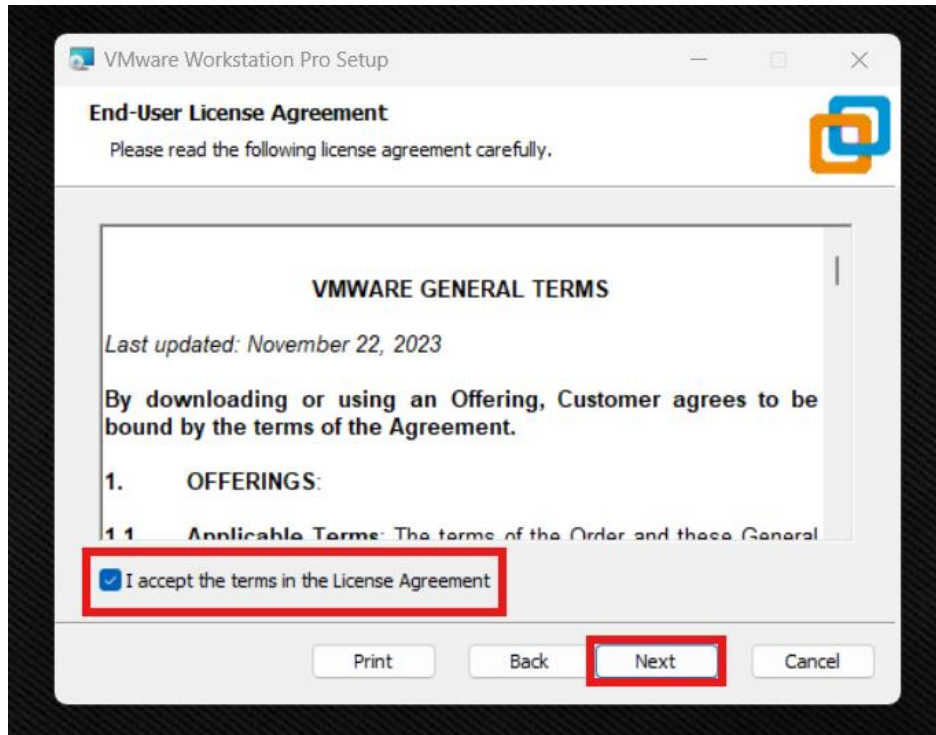


Figure 19 - Download of VMware Workstation Pro

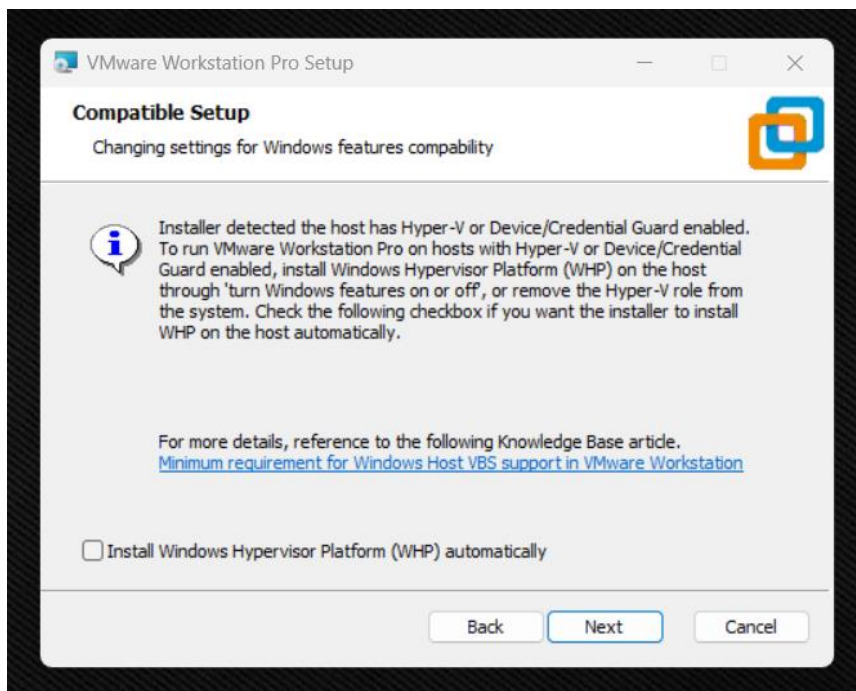
17 - After the executable has been downloaded, run it to start the installation. The window that will appear is this. Select "Next"



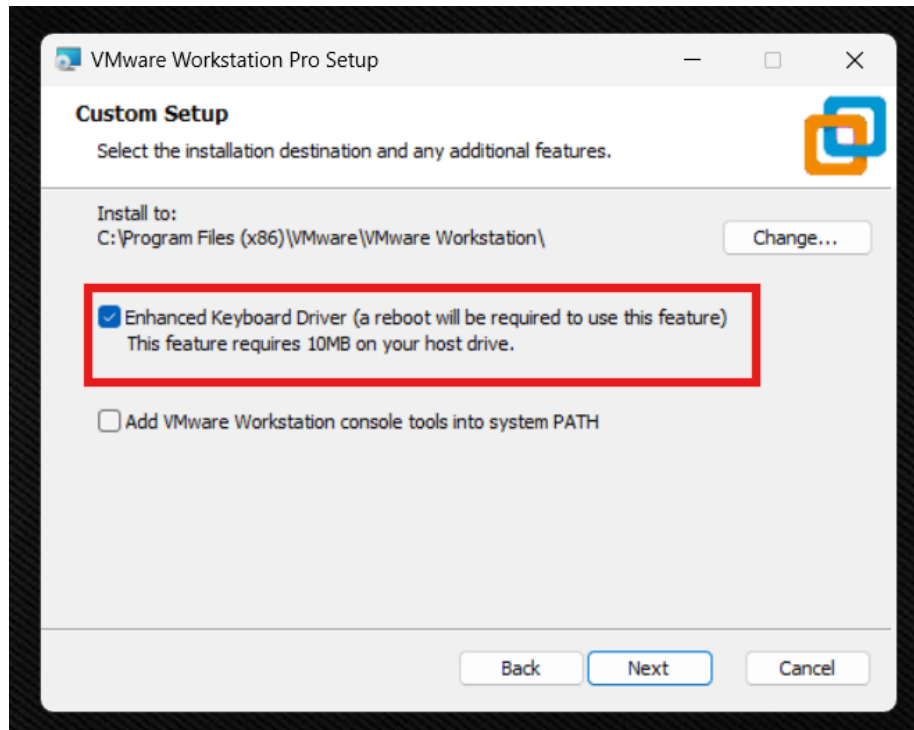
18 - After clicking on "Next" the program license will appear, accept the terms and continue



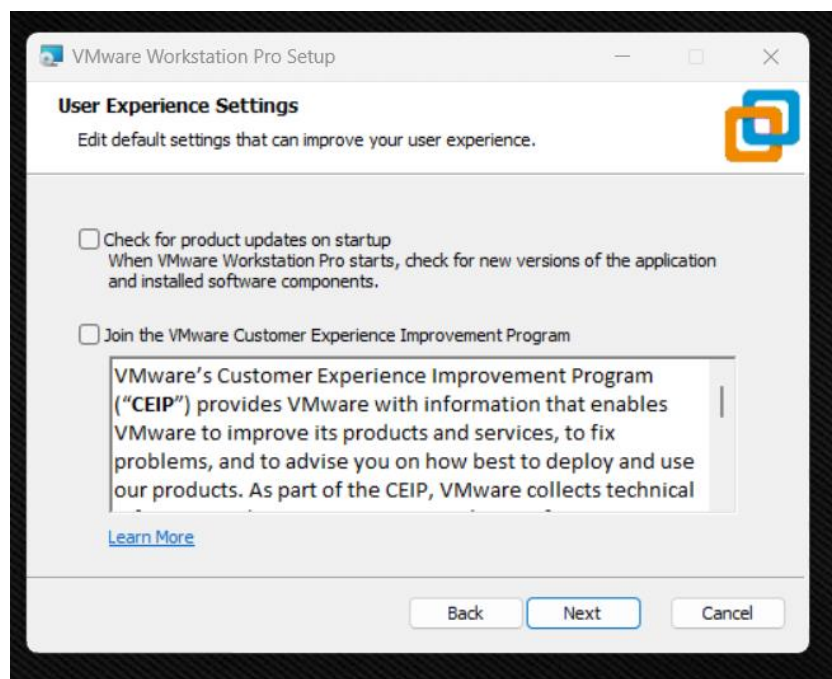
19 - After accepting the program's license terms, it will ask if you want to install the Windows Hypervisor Platform, hacktux doesn't require this feature so it's up to you whether you want to install it or not.



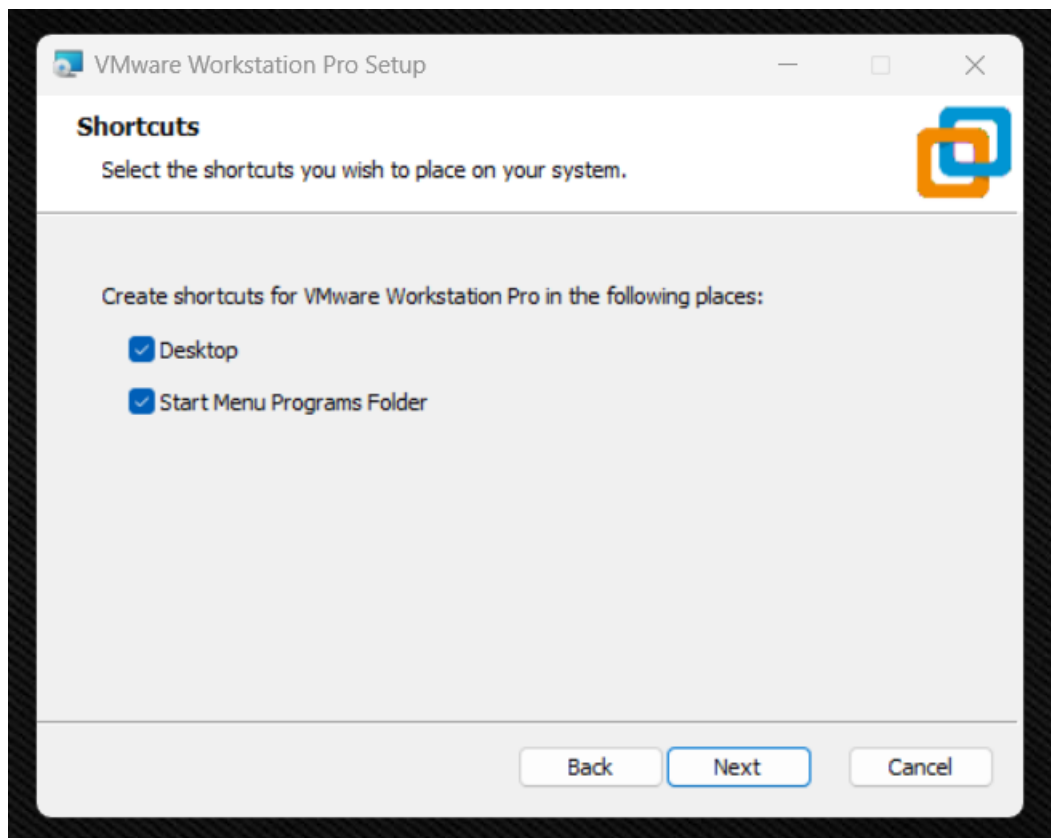
20 - In the next window it is possible to customize some settings, the first is to change the path where VMware Workstation Pro will be installed, it is up to the user to decide, then the Keyboard Driver feature must be accepted, the last option does not need to be selected, it is up to the user's discretion



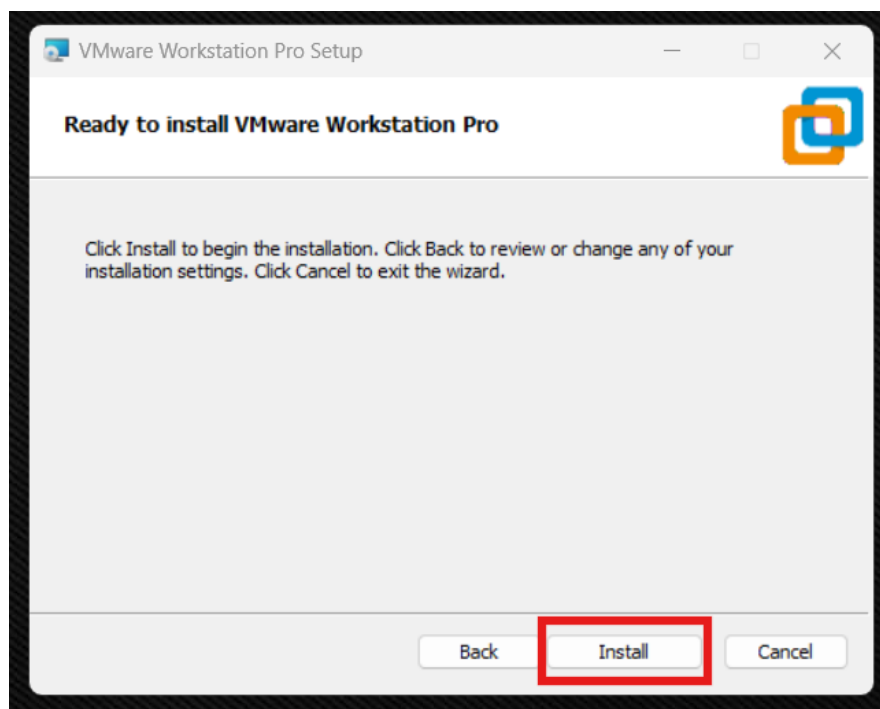
21 - The next settings are up to the user's preference, by default we advise you to disable both of them



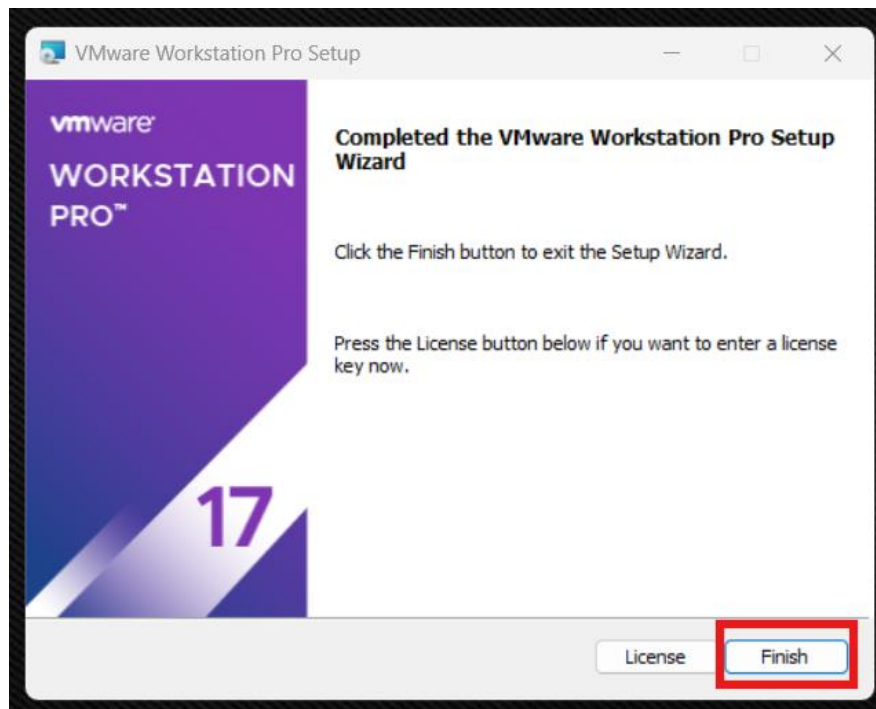
22 - The next settings are up to the user's preference



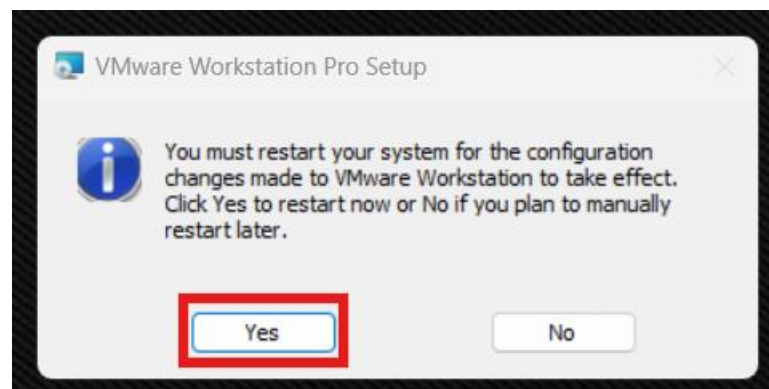
23 - After all these steps in the next window just click Install and the installation process will begin



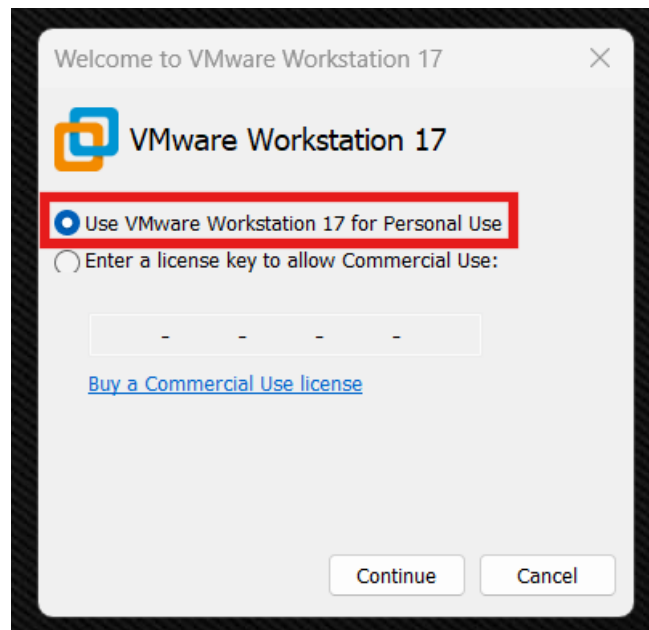
23 - Once the installation has been successfully completed, simply press the "Finish" button



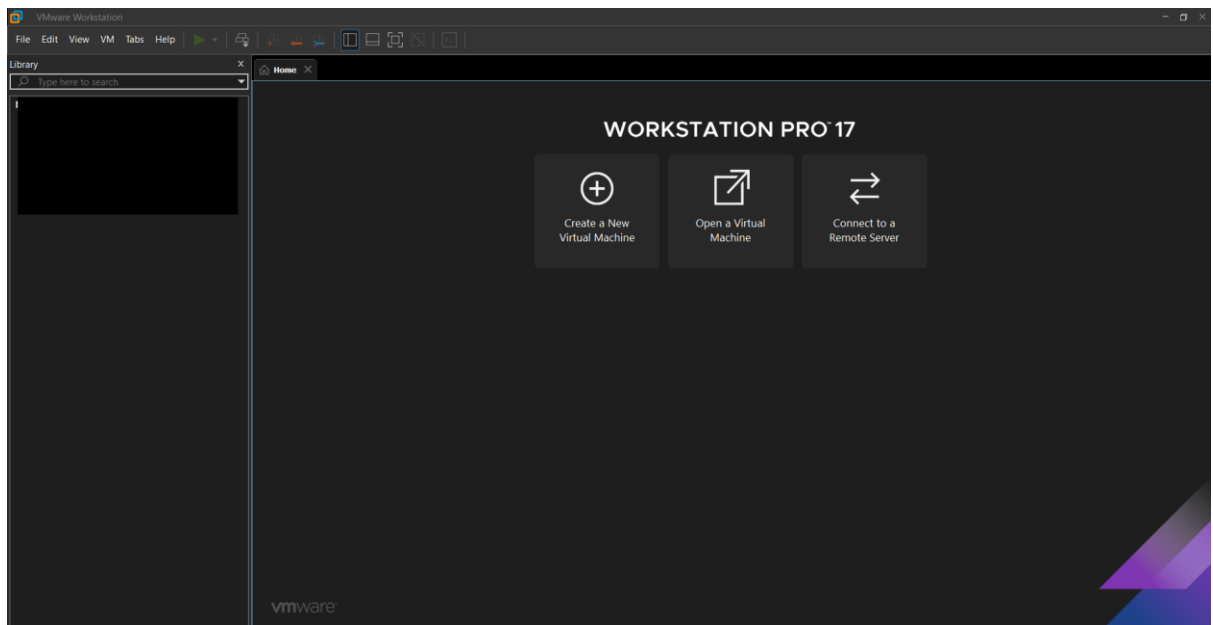
24 - After pressing the "Finish" button, a window will appear asking you to restart your computer. Press "Yes" so that the settings are made and VMware Workstation Pro is ready to use.



25 - After restarting your computer, just start vmware workstation, when the application opens, the following window will appear, then select the option "Use VMware Workstation 17 for Personal Use"



26 - After that, just press the "Finish" button and VMware Workstation Pro has been successfully installed and you can start using it.



2.3. Installation of Hacktux Application

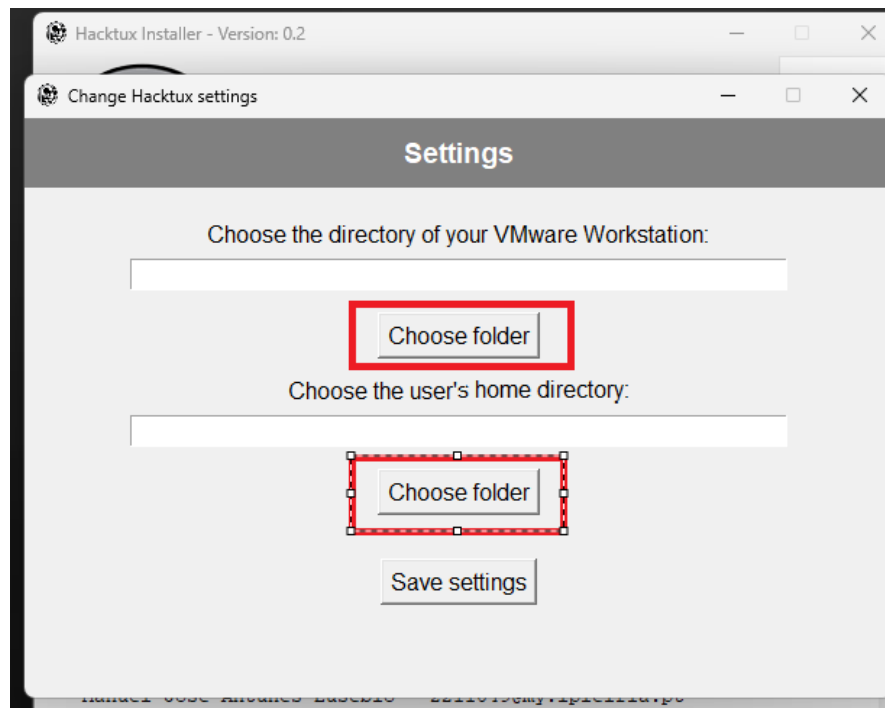
1-To install all the dependencies and configure the virtual machines and their association with VMware, you need to use our installer to make it more practical.



1.1 – If you don't have the default vmware and user folders, you should change them to where they are on your computer using the change settings menu.



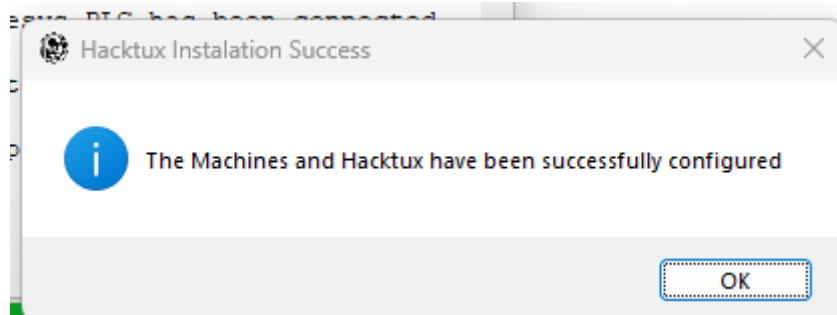
- 1.2 – To make this choice, simply click on the choose folder button below the blank text-box, or you can choose to type the path manually in the text-box above the button.



- 2- To carry out the installation, simply press the 'Accept' button after reading all the information in the text box.



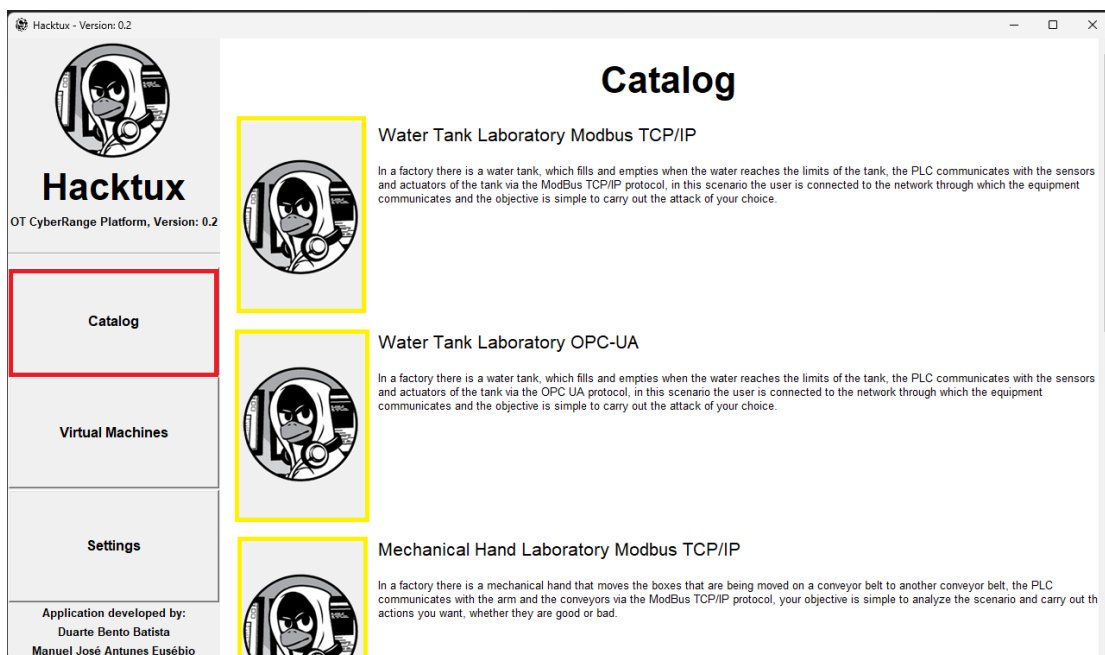
3- Once you've finished, click on the Ok button to exit the interaction menu.



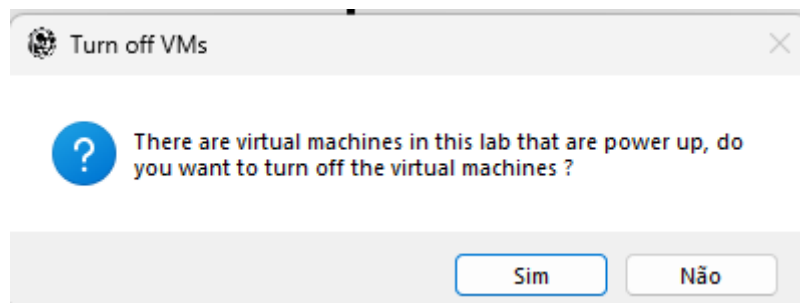
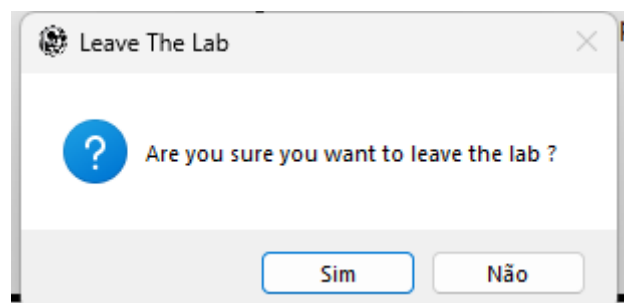
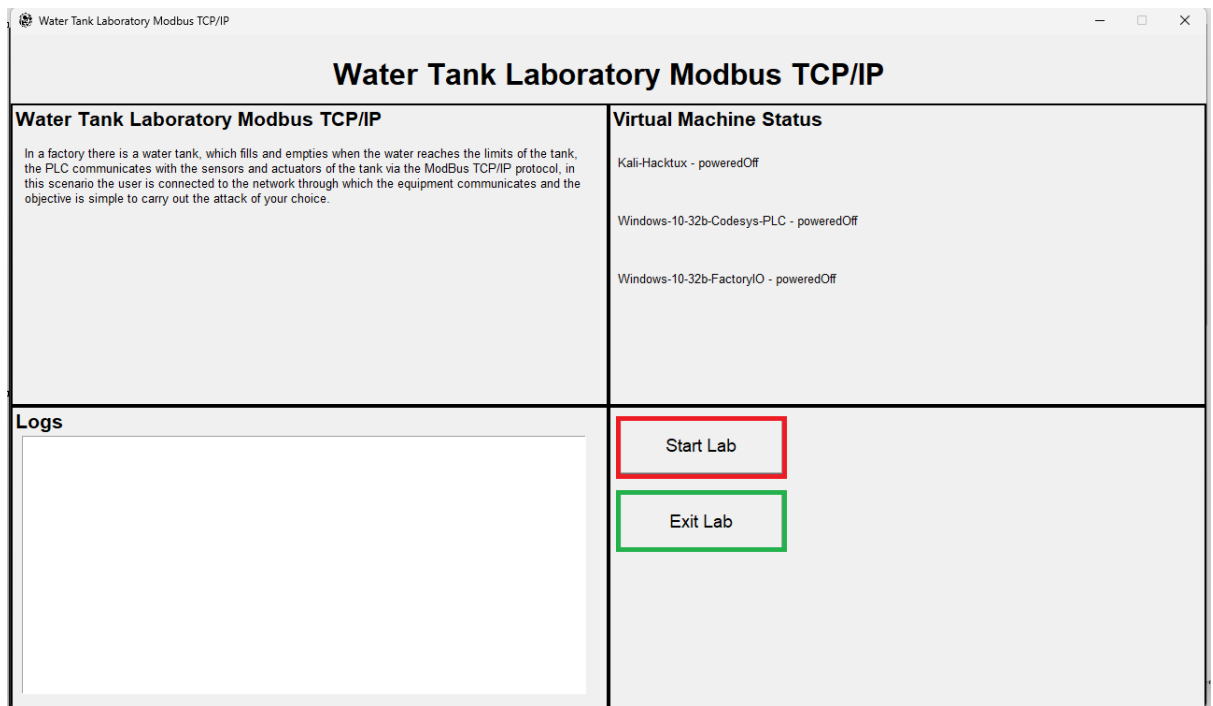
3. User Interface Overview

3.1. Hacktux Application Catalog

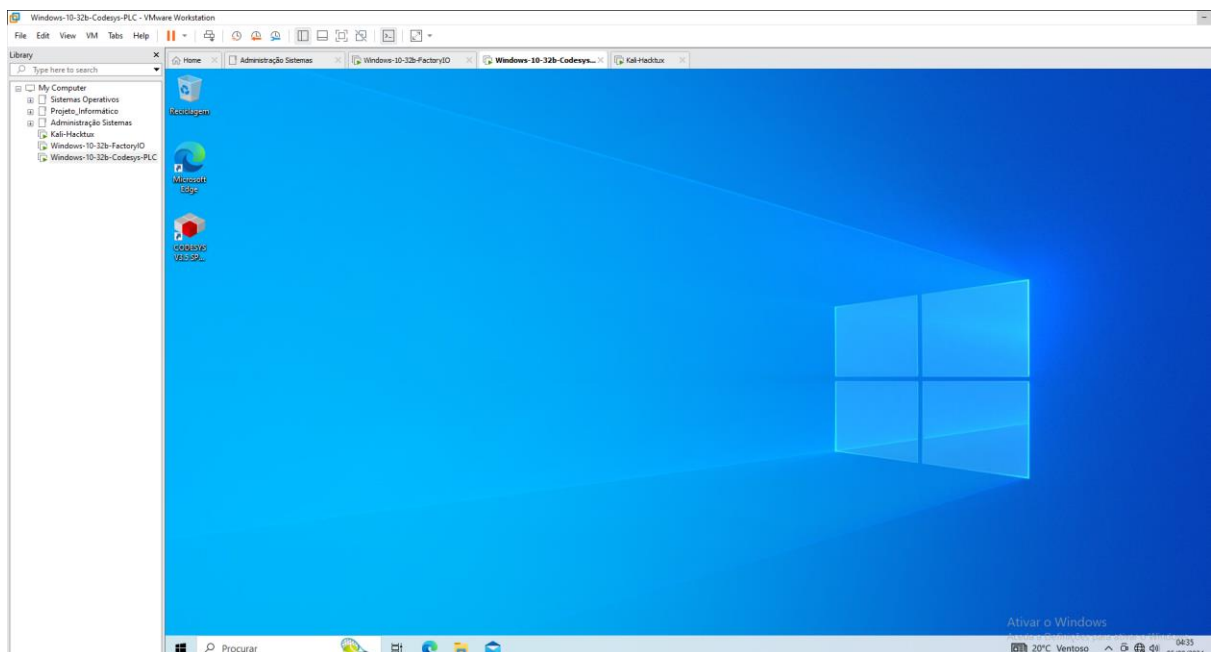
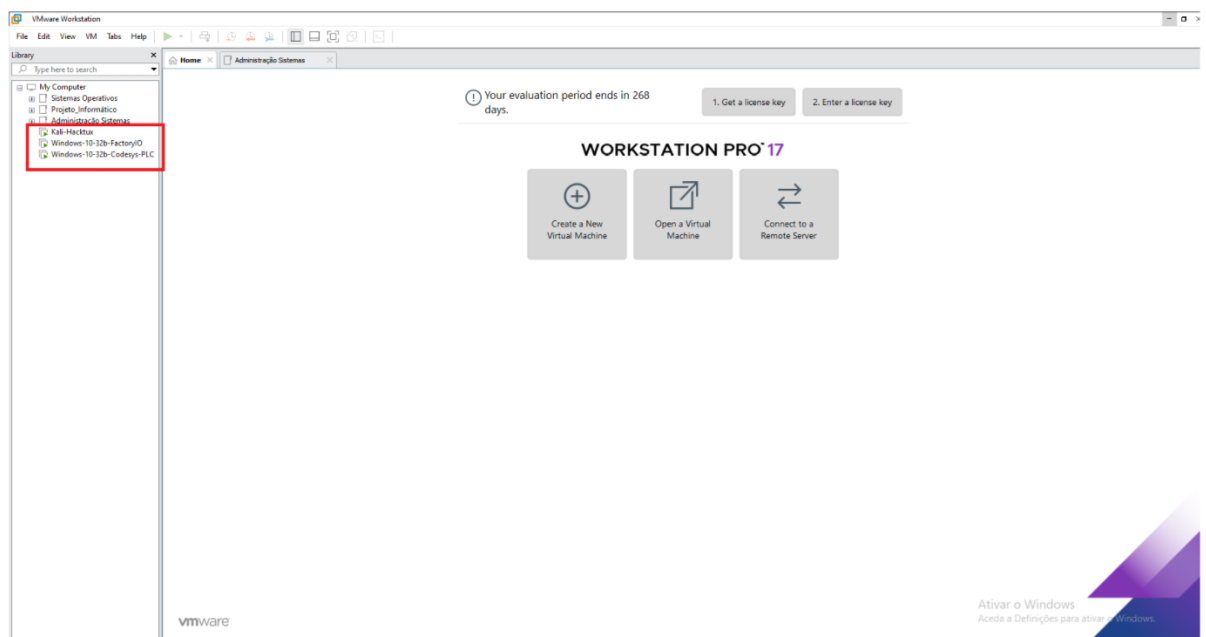
1- After installing all the dependencies, simply enter the application, in the Catalogue menu, highlighted in red, you can select the labs you want to do by clicking on the image in front of it, highlighted in yellow.



2- Once you have entered the desired lab, simply select the Start Lab option, shown in red, to begin the process of entering the labs, which takes around 5 minutes. If you want to leave, simply click on the Exit Lab button, shown in green, accept the confirmation message and choose whether you want to switch off the machines or not.

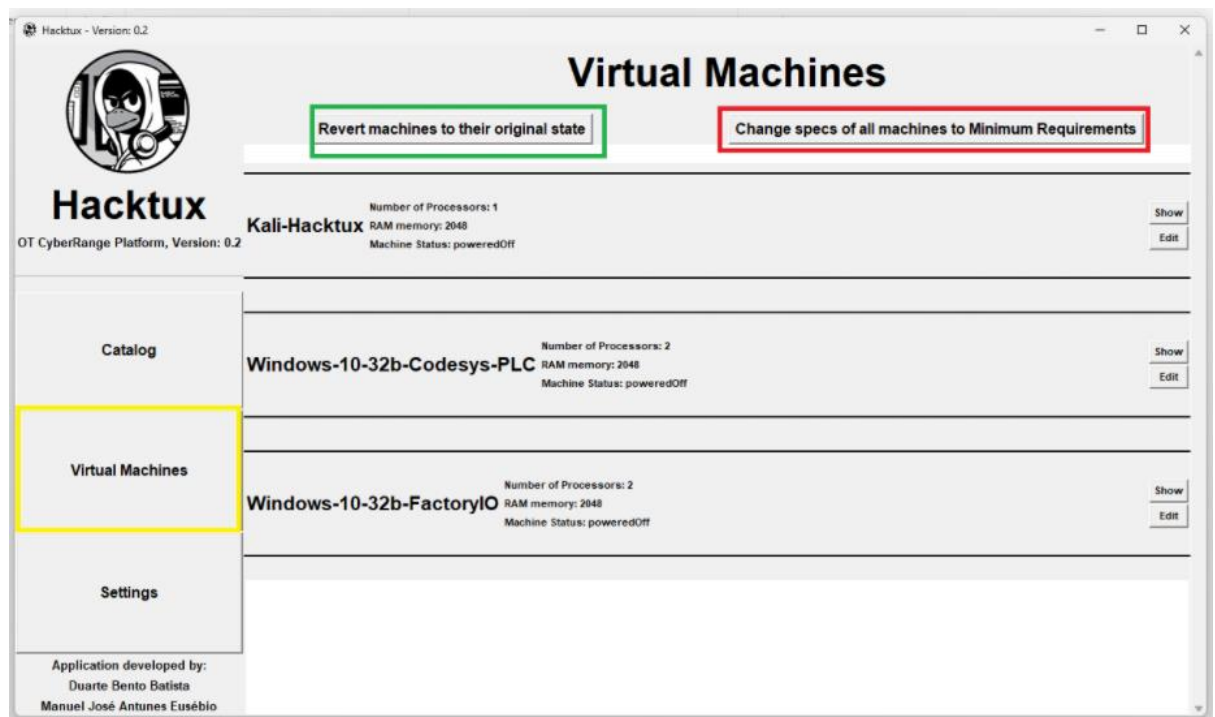


3- After initialising the lab, simply select one of the 3 machines in VMware with the green play symbol in the left-hand sidebar, as shown in red in the following image; once selected, simply use the machine normally as if it were your computer.

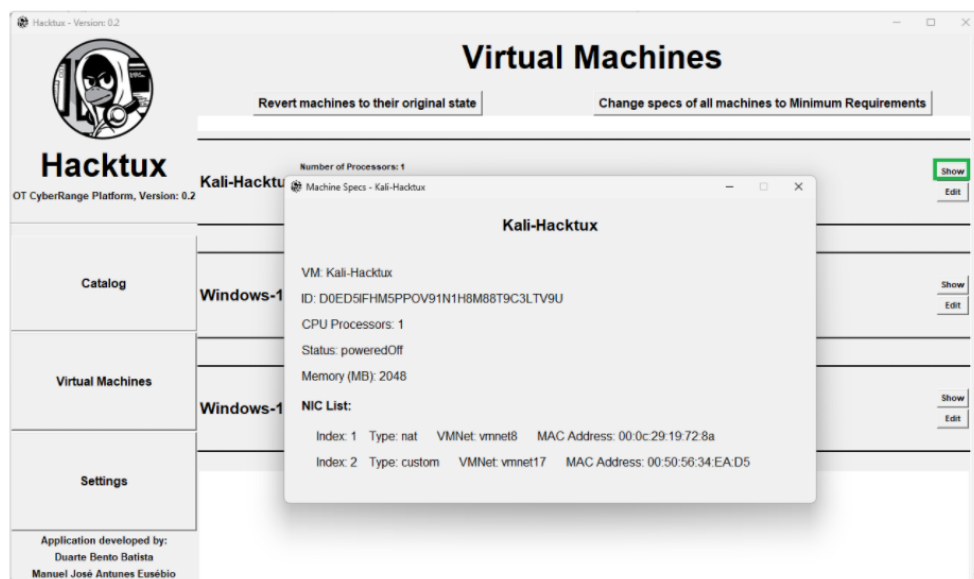


3.2. Hacktux Aplicattion Virtual machines

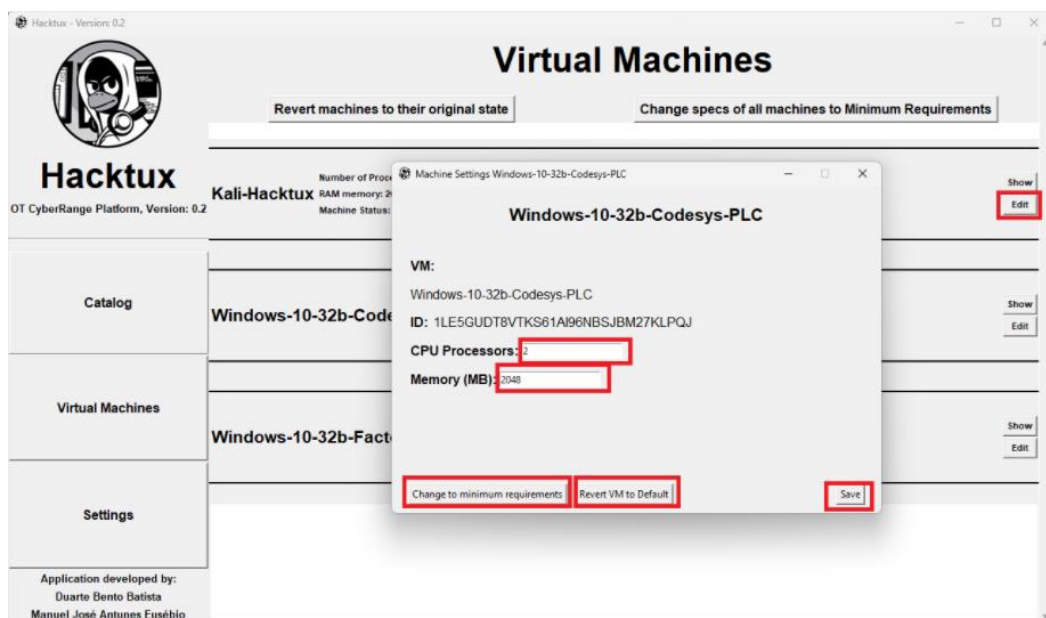
1– In the Virtual Machines menu, shown in yellow, you can revert all of them to their original state, shown in green, and set all of them to their minimum requirements, shown in red, recommended for weaker computers.



2– The show option, highlighted in green, shows all the characteristics of the virtual machines.



3- In the edit option, highlighted in red, you can edit the characteristics of each of the virtual machines, only do this if you know what you're doing. In the bottom left-hand corner you can revert the machine to its original state or set it to the minimum requirements. In the squares highlighted in red you can enter the number of CPUs and the amount of RAM in MB and then click the Save button in the bottom right-hand corner, changing the machine's characteristics.



3.3. Hacktux Application Settings

1- In the settings menu you can change the default folders for both the user and VMware Workstation Pro.

